ITERON

Inspired by: Robert Axelrod's Iterated Prisoner's Dilemma Tournament

Organized by:
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Abstract

This document outlines the official rules, submission guidelines, and scoring criteria for the algorithmic game theory tournament. Participants are advised to read all sections carefully and follow the instructions.

General Information

Event Description:	A round-robin computational tournament designed to evaluate strategies across a variety of strategic environments using cumulative payoffs.
Medium:	Online
Eligibility:	Open to all students (School, Undergraduate, Postgraduate, and PhD, associated with some university).

Contents

1 Introduction

Welcome to the Iteron. This tournament is inspired by Axelrod's Iterated Prisoner's Dilemma tournament. Participants are required to submit code-based strategies to compete against one another. Unlike conventional tournaments, this event tests the robustness of strategies across varying environments.

2 Submission Guidelines

2.1 Strategy Originality And Structure

- All submitted strategies must be **original and unique**. Plagiarism of existing open-source strategies without significant modification or attribution is strictly prohibited.
- References will be provided by the organizers. Participants must ensure their strategies are distinct from these reference implementations.

2.2 Formatting and Templates

- Strategies must be submitted using the **official template** provided by the organizers. provided by the organizers.
- Submissions in any other format or structure will be rejected. Participants may be asked to resubmit in the correct format or face immediate disqualification.

2.3 Testing

- To prevent runtime errors during the actual tournament, the organizers will release a **Sample Strategy**.
- You are required to test your code against this sample strategy before submission.
- Detailed instructions on how to perform this pre-submission test will be issued shortly.

2.4 Technical Constraints

- **Machine Learning:** The use of Machine Learning (ML) within strategies is expected to be **limited**. Strategies relying heavily on computationally expensive models may be subject to review regarding time-complexity limits.

3 Tournament Mechanics

3.1 Structure

- The tournament will be conducted in a **Round Robin** format.
- Every submitted strategy will play against:

1. Every other opponent's strategy.
2. A copy of itself.

3.2 Match Duration

- To preserve the game-theoretic essence of the "Shadow of the Future," the exact number of rounds per match **will not be revealed** to the participants.
- Strategies should be designed to handle an indefinite time horizon.

3.3 Iteration

- The entire tournament will be iterated more than once. This is to ensure statistical significance, remove noise, and ensure fair observations of strategy performance.

4 The Multi-Environment System

Note: This is the defining feature of this tournament.

- Unlike the conventional Axelrod's tournament which uses a static payoff matrix, this tournament will be conducted across **multiple environments**.
- The organizers will systematically vary the payoff matrices (the rewards for Cooperation and Defection) between tournament runs.
- Strategies must be robust enough to adapt to changes in incentives, rather than being over-fitted to a single payoff structure.
- The description of payoffs will be mentioned on the website.

5 Scoring and Ranking

5.1 Winner Determination

- The final winner will be determined based on the **highest average score**.
- This score is calculated by aggregating the points obtained across **all** the different payoff matrices and environments.

5.2 Aggregate Performance Rule

- **Consistency is key.** Even if a strategy places 1^{st} in k out of n specific matrix environments, it will receive no special consideration if its overall average drops due to poor performance in other environments.
- Final ranking is based *solely* on the overall aggregated performance. No separate claims or titles will be awarded for individual sub-tournament wins.

The Turing Club reserves the right to modify these rules to ensure the smooth operation of the Science Fest event. Updates will be communicated via official channels.